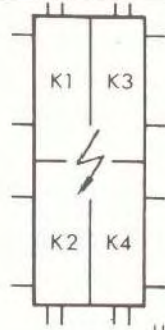
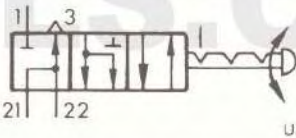
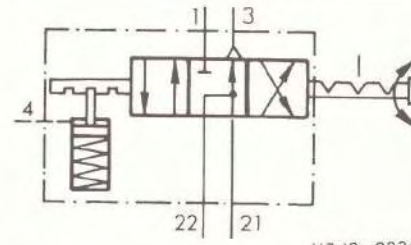
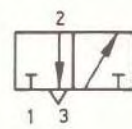
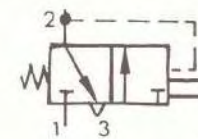


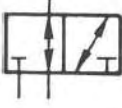
No.	Designation	Symbol
Electrical unit		
31.1	4-channel ABS	 UZ43-0233

32 Directional control valves with 3 positions

32.1	4/3-way valve, mechanically actuated	 UZ43-0177
32.2	4/3-way valve, mechanically actuated	 UZ43-0234
32.3	4/3-way valve, mechanically actuated	 UZ43-0235

33 Directional control valves with 2 positions

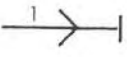
33.1	3/2-way valve	
33.2	Safety valve (Sweden)	 UZ43-0178

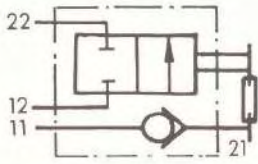
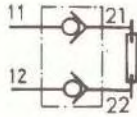
No.	Designation	Symbol
33.3	3/2-way valve	 UZ43-0236

34 Reservoir (hydraulic)

34.1	— With connection at bottom	 UZ43-0179
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35 Coupling heads

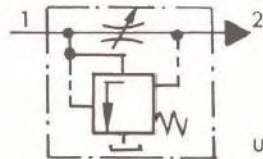
35.1	— For truck brakes (EEC) (polarized without shutoff valve)	 UZ43-0249
35.6	— For dual-line brakes (Pam type) (brakes and supply, without shutoff valve)	
35.0	— For special version (without shutoff valve)	
35.2	— For tractive unit supply (EEC) (polarized with shutoff valve)	 UZ43-0248
35.3	— For tractive unit brakes (EEC) (polarized with shutoff valve)	
35.5	— For single-line brake (with shutoff valve)	
35.7	— For special version pneumatics, hydraulics (e.g. for tipping platform, quick coupling)	

No.	Designation	Symbol
35.9	– For single-line brakes (with shutoff valve and 2/2-way valve)	
35.8	– Double for dual-line brakes (e.g. Italy, Sweden, Norway)	 UZ 43-0250
35.4	– For truck supply (EEC) (polarized with shutoff valve)	 UZ 43-0180/1

36 Filters

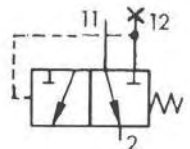
36.1	Line filter	 UZ 43-0190
36.2	– With automatic water separator	 UZ 43-0237
36.3	Intake filter	 UZ 43-0238

37 Throttle valves

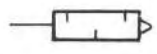
37.1	Throttle valve	
37.2	– Variable	
37.3	Flow control valve (shunt-connected)	 UZ 43-0181

No.	Designation	Symbol
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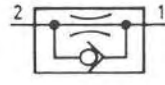
38 Pressure connection points

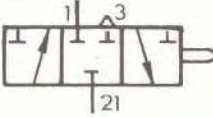
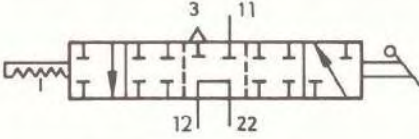
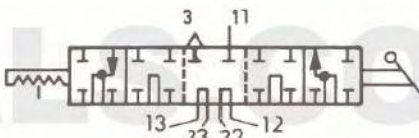
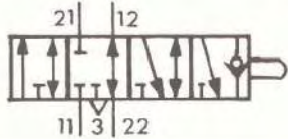
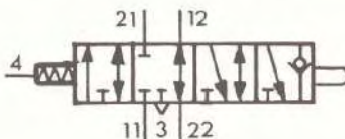
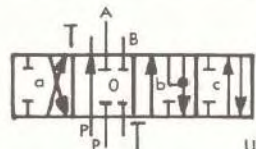
38.1	Test connection Vg 8 (for pneumatics DIN 74 326)	
38.2	Test connection M 16 x 1.5 (for pneumatics DIN 74 326)	
38.3	Test connection M 18 x 1.5 (für hydraulics DIN 72 237)	
38.4	Test valve for pressure-controlled ALB	 UZ 43-0182
38.5	Charging valve, emergency release connection for spring-loaded cylinder	 UZ 43-0239

39 Bleed point

39.1	Bleed point for hydraulics	 UZ 43-0183
39.2	Silencer	 UZ 43-0240

40 One-way restrictors

40.1	One-way restrictor	
40.2	– Adjustable	 UZ 43-0184

No.	Designation	Symbol
41 Directional control valve with 3 positions		
41.1	Air spring valve	 UZ 43-0242
42 Directional control valves with 4 positions		
42.1	Relay valve, single-circuit for demountable bodies (4/4-way valve)	
42.2	Relay valve, dual-circuit for demountable bodies (6/4-way valve)	
42.3	Air spring valve with vertical limitation (4/4-way valve)	
42.4	Air spring valve with vertical limitation and zero position adjustment (4/4-way valve)	
42.5	5/4-way valve	
42.6	6/4-way valve	 UZ 43-0186

No.	Designation	Symbol
42.7	Relay valve, dual-circuit with closed position	 UZ 43-0245
42.8	Relay valve, 3-circuit	 UZ 43-0243

43 Air spring bellows

43.1	Tubular bellows	
43.2	Pleated bellows	 UZ 43-0187

45 Directional control valves with 2 positions

45.1	2/2-way valve	
45.2	4/2-way valve	
45.3	2/2-way valve with throttle	 UZ 43-0188

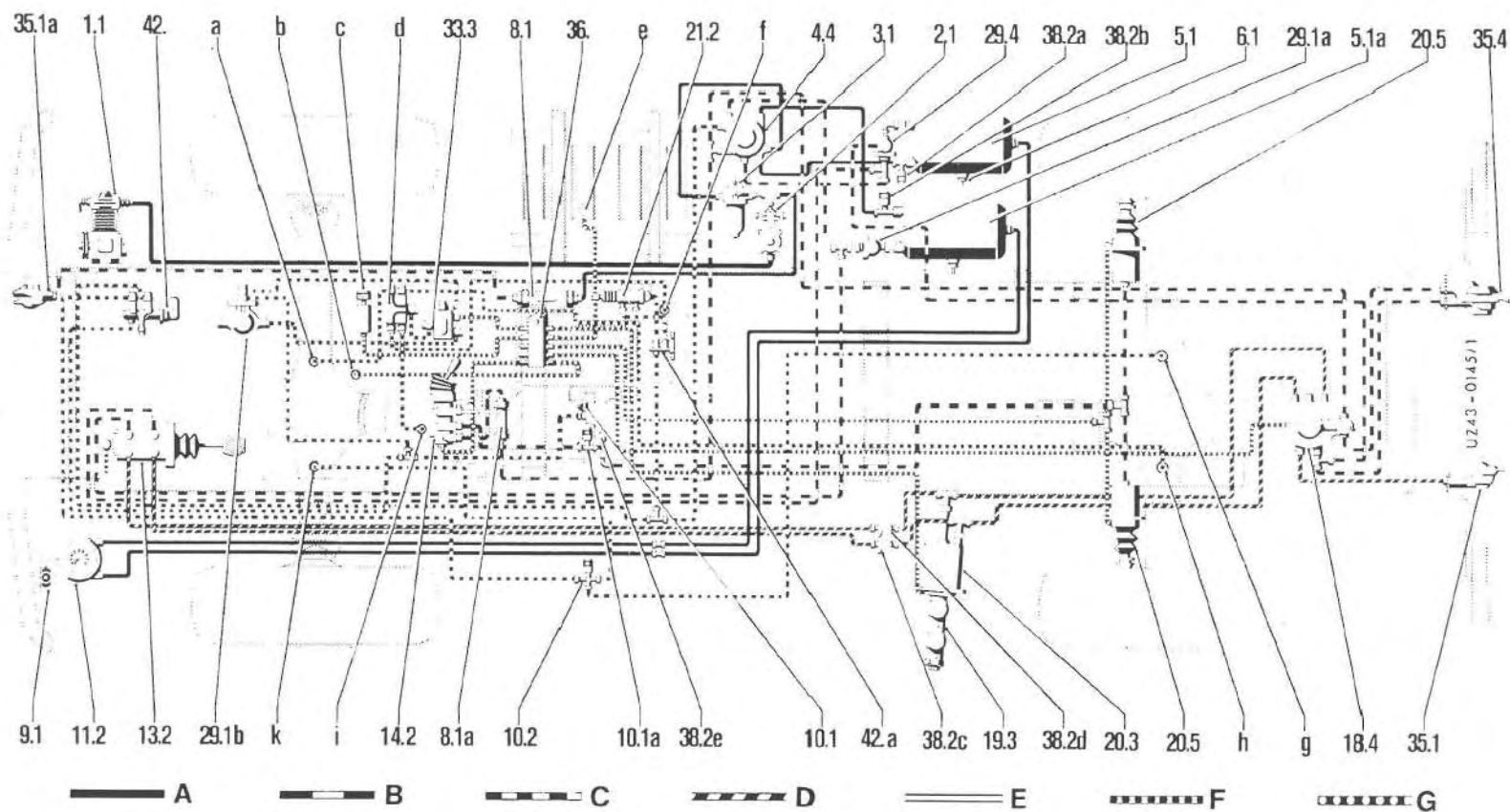
No.	Designation	Symbol
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46 Linkages

46.1	Adjustable	 UZ 43 – 0244
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Brake Diagram (Special Vehicle 435.115)

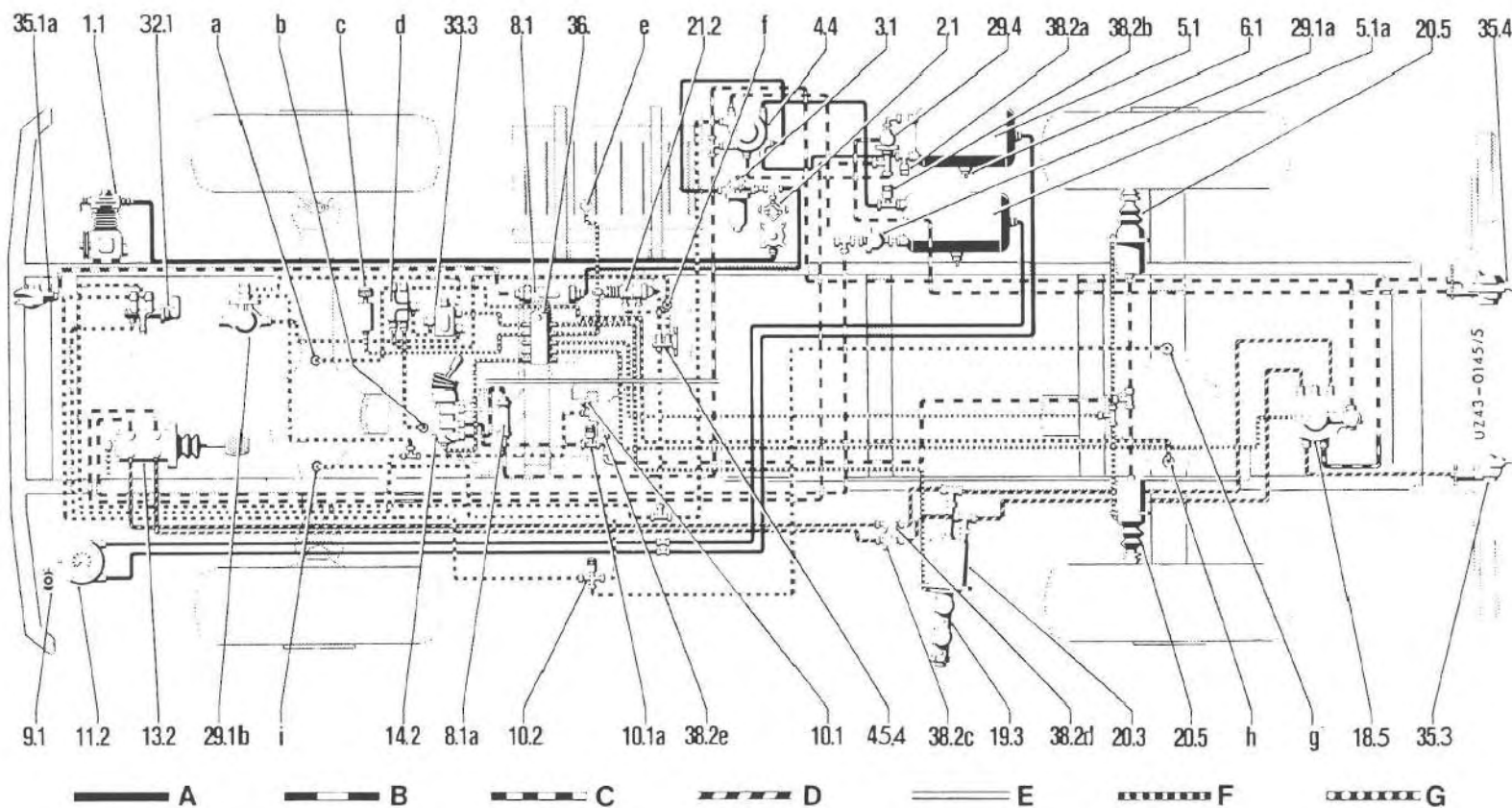


Dual-circuit air brake system, spring-loaded parking brake and dual-line trailer brake system. To chassis end No. 105 688

Key to 1.6/1

- | | | | |
|---|-----------------------------------|---|---|
| A | High pressure supply 18 bar | a | Connection, front axle air admittance and escape |
| B | Supply brake 9,5 bar or 7,0 bar | b | Connection, clutch casing air escape |
| C | Auxiliary load 9,5 bar or 7,0 bar | c | Starting pilot vent |
| D | Brake control pressure | d | Compressed air distributor |
| E | Trailer brake test pressure | e | Fuel tank vent |
| F | Air admission | f | Connection, transmission air admission and escape |
| G | Central vent | g | Connection, differential rear axle |
| | | h | Connection, rear axle air admission and escape |
| | | i | Connection, PTO transmission air admission and escape |
| | | k | Connection, differential lock front axle |
-
- | | |
|-------|---|
| 1.1 | Air compressor |
| 2.1 | Pressure regulator with tyre inflating port |
| 3.1 | Antifreeze unit |
| 4.4 | Four-circuit protection valve |
| 5.1 | Air tank (bottom) |
| 5.1a | Air tank (top) |
| 6.1 | Drain valve |
| 8.1 | Non-return valve, charging line |
| 8.1a | Non-return valve, spring-loaded cylinder |
| 9.1 | Warning lamp supply pressure |
| 10.1 | Electrical pressure switch 4-wheel brake system |
| 10.1a | Electrical pressure switch parking brake |
| 10.2 | Electrical pressure switch differential lock |
| 11.2 | Double pressure gauge (supply) |
| 13.2 | Service brake valve (dual-circuit) |
| 14.2 | Parking brake valve with brake test point |
| 18.4 | Trailer control valve (2-line) with breakaway valve |
| 19.3 | Master brake cylinder (dual-circuit) |
| 20.3 | Preload cylinder (dual-circuit) |
| 20.5 | Spring-loaded parking brake |
| 21.2 | Shift cylinder, planetary gear |
| 29.4 | Pressure reducing valve, can be switched over for trailer operation, truck operation 9,5 bar, trailer operation 7,0 bar circuit 1 |
| 29.1a | Pressure reducing valve circuit 2 |
| 29.1b | Pressure reducing valve air admission |
| 33.3 | 3/2-way valve air admission |
| 35.1 | Coupling head brakes |
| 35.1a | Coupling head charging connection |
| 35.4 | Coupling head supply with shutoff valve |
| 36. | Central air escape filter |
| 38.2a | Test connection supply pressure circuit 1 |
| 38.2b | Test connection supply pressure circuit 2 |
| 38.2c | Test connection brake pressure circuit 1
9,5 bar truck operation 7,0 bar trailer operation |
| 38.2d | Test connection brake pressure circuit 2 |
| 38.2e | Test connection spring-loaded cylinder |
| 42. | Switchover valve 4-wheel differential lock |
| 42a | Switchover valve planetary gear |

Brake Diagram (Special Vehicle 435.115)



Dual-circuit air brake system, spring-loaded
parking brake and two-line trailer brake system.
From Chassis End No. 105 689

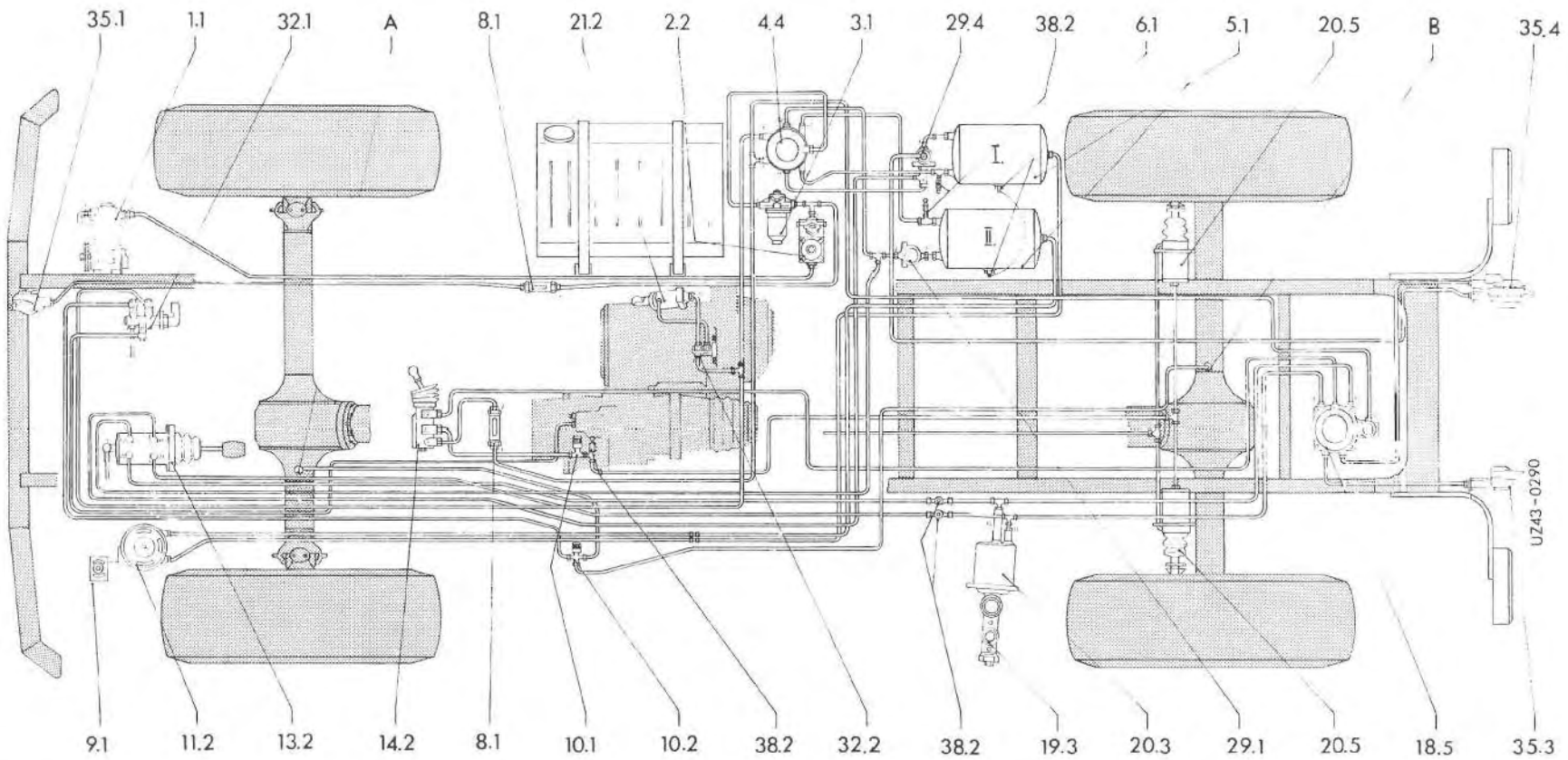
Key to 1.6/3

- A = High pressure supply 18 bar
- B = Supply, brakes 9,5 bar and 7,0 bar
- C = Auxiliary load 9,5 bar and 7,0 bar
- D = Brake control pressure
- E = Trailer brake test pressure
- F = Air admission
- G = Central vent

- a Connection, front axle air admission and escape
- b Connection, PTO transmission air admission and escape
- c Starting pilot vent
- d Compressed air distributor
- e Fuel tank vent
- f Connection, transmission air admission and escape
- g Connection, differential lock rear axle
- h Connection, rear axle air admission and escape
- i Connection, differential lock front axle

- 1.1 Air compressor
- 2.1 Pressure regulator with tyre inflating port
- 3.1 Antifreeze unit
- 4.4 Four-circuit protection valve
- 5.1 Air tank (bottom)
- 5.1a Air tank (top)
- 6.1 Drain valve
- 8.1 Non-return valve, charging line
- 8.1a Non-return valve, spring-loaded cylinder
- 9.1 Warning lamp, supply pressure
- 10.1 Electrical pressure switch 4-wheel brake system
- 10.1a Electrical pressure switch parking brake
- 10.2 Electrical pressure switch differential lock
- 11.2 Double pressure gauge (supply)
- 13.2 Service brake valve (dual-circuit)
- 14.2 Parking brake valve with brake test point
- 18.5 Trailer control valve (2-line) with breakaway valve
- 19.3 Master brake cylinder (dual-circuit)
- 20.3 Preload cylinder (dual-circuit)
- 20.5 Spring-loaded parking brake
- 21.2 Shift cylinder planetary gear
- 29.4 Pressure reducing valve can be switched over to trailer operation 9,5 bar, trailer operation 7,0 bar circuit 1
- 29.1a Pressure reducing valve circuit 2
- 29.1b Pressure reducing valve air admission
- 33.3 3/2-way valve air admission
- 35.1a Coupling head charging connection
- 35.3 Coupling head break
- 35.4 Coupling head supply with shut-off valve
- 36. Central vent filter
- 38.2a Test connections supply pressure circuit 1
- 38.2b Test connections supply pressure circuit 2
- 38.2c Test connections brake pressure circuit 1
9,5 bar truck operation 7,0 bar trailer operation
- 38.2d Test connection brake pressure circuit 2
- 38.2e Test connection spring-loaded cylinder
- 42. Switchover valve 4-wheel differential lock
- 42a Switchover valve planetary gear

Brake Diagram (435.115/117)



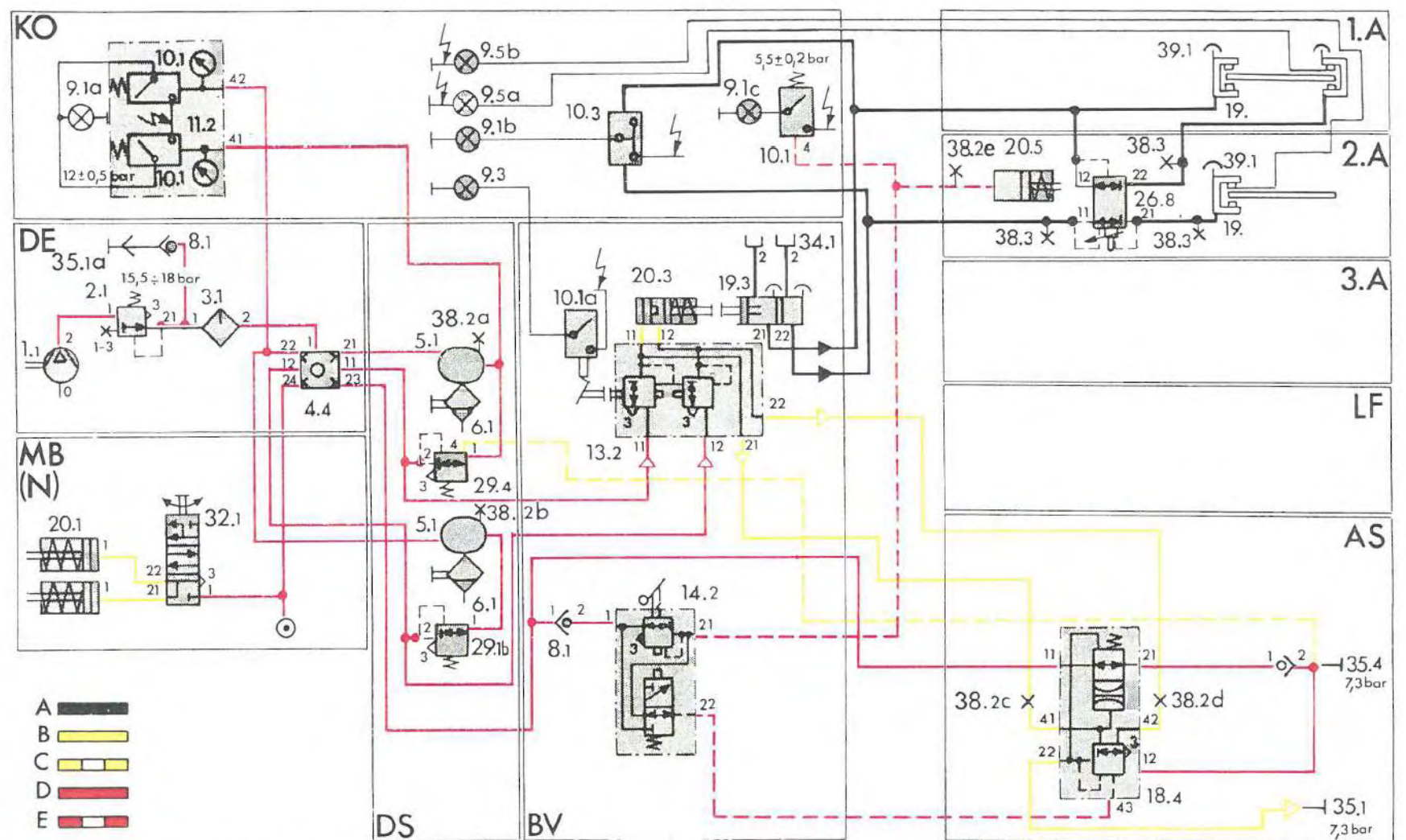
Dual-circuit air brake system and two-line trailer brake system, spring-loaded parking brake and auxiliary load

Key to 1.6/5

- | | |
|---|--|
| 1.1 Single cylinder air compressor | 20.3 Preload cylinder, dual circuit |
| 2.2 Pressure regulator | 20.5 Spring-loaded brake cylinder |
| 3.1 Antifreeze unit | 21.2 Shift cylinder |
| 4.4 Four-circuit protection valve | 29.1 Pressure reducing valve |
| 5.1 Air tank | 29.4 Pressure reducing valve, switchover |
| 6.1 Drain valve (water separator) | 32.1 Switchover valve, four-wheel drive differential |
| 8.1 Non-return valve | 32.2 locks |
| 9.1 Warning lamp, red | 35.1 4/2-way valve |
| 10.1 Pressure switch, parking brake | 35.3 Charging line for front supply |
| 10.2 Pressure switch, differential lock | 35.4 Coupling head, brakes |
| 11.2 Instrument cluster | 38.2 Coupling head, supply |
| 13.2 Service brake valve | 38.2 Test connections, brake pressure circuit I and circuit II |
| 14.2 Handbrake valve | |
| 18.5 Trailer control valve with breakaway valve | A Connection, differential lock, front |
| 19.3 Master brake cylinder | B Connection, differential lock, rear |

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Functional Diagram (Special Vehicle 435.115) to chassis end No. 084 388



Dual-circuit hydraulic brake system with dual-circuit air brake system. ALB on rear axle and 1/2 front axle, spring-loaded parking brake and dual-line trailer brake system.

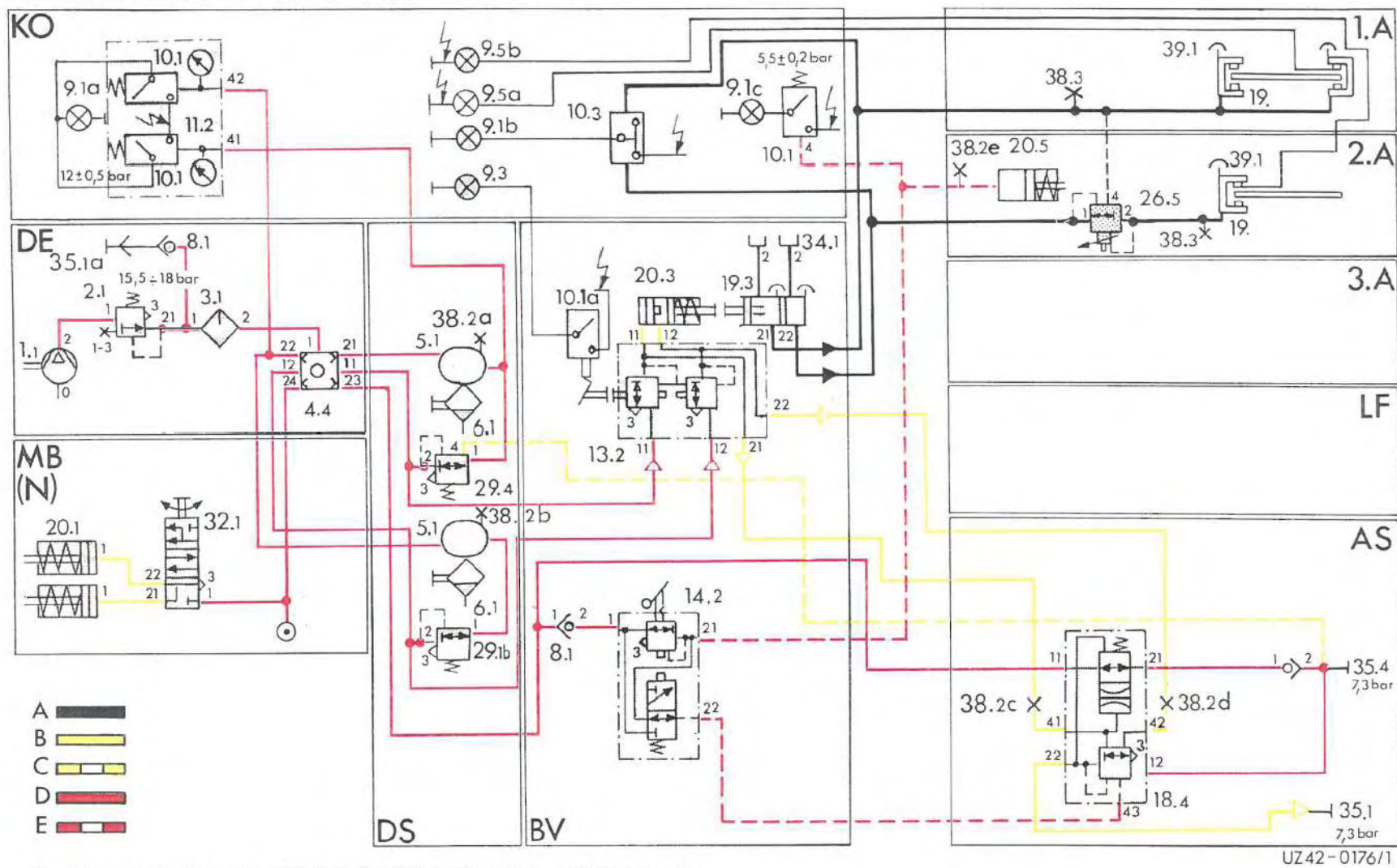
UZ 42- 0075/2

Key to 1.7/1

- A Hydraulics
- B Brake pressure
- C Brake control pressure
- D Supply pressure
- E Supply control pressure

- 1.1 Air compressor
- 2.1 Pressure regulator with tyre inflating port
- 3.1 Antifreeze unit
- 4.4 Four-circuit protection valve
- 5.1 Air tank circuit 1
- 5.1a Air tank circuit 2
- 6.1 Drain valve
- 8.1 Non-return valve
- 9.1a Warning lamp, supply pressure
- 9.1b Indicator lamp, pressure difference
- 9.1c Indicator lamp, parking brake
- 9.3 Stoplight
- 9.5a Indicator lamp wear indicator FA
- 9.5b Indicator lamp wear indicator RA
- 10.1 Electrical pressure switch supply pressure
- 10.1a Electrical pressure switch stoplight
- 10.3 Electrical pressure difference warning switch
- 11.2 Double pressure gauge
- 13.2 Service brake valve (dual-circuit)
- 14.2 Parking brake valve with brake test point
- 18.4 Trailer control valve (2-line) with breakaway valve
- 19. Fixed caliper
- 19.3 Master brake cylinder (dual-circuit)
- 20.1 Piston cylinder 4-wheel drive differential lock
- 20.3 Preload cylinder (dual-circuit)
- 20.5 Spring-loaded parking brake
- 26.8 Automatic brake pressure modulator (ALB)
- 29.1 Pressure reducing valve
- 29.4 Pressure reducing valve with pneumatic control
- 32.1 Switchover valve 4-wheel differential lock
- 34.1 Equalizing tank
- 35.1 Coupling head brake
- 35.1a Coupling head front charging line
- 35.4 Coupling head supply
- 38.2a Test connection circuit 1
- 38.2b Test connection circuit 2
- 38.2c Test connection circuit 1
- 38.2d Test connection circuit 2
- 38.2e Test connection, spring loaded cylinder
- 38.3 Test connection ALB modulator
- 39.1 Bleeder valve fixed caliper

Functional Diagram (Special Vehicle 435.115) to chassis end No. 105 688



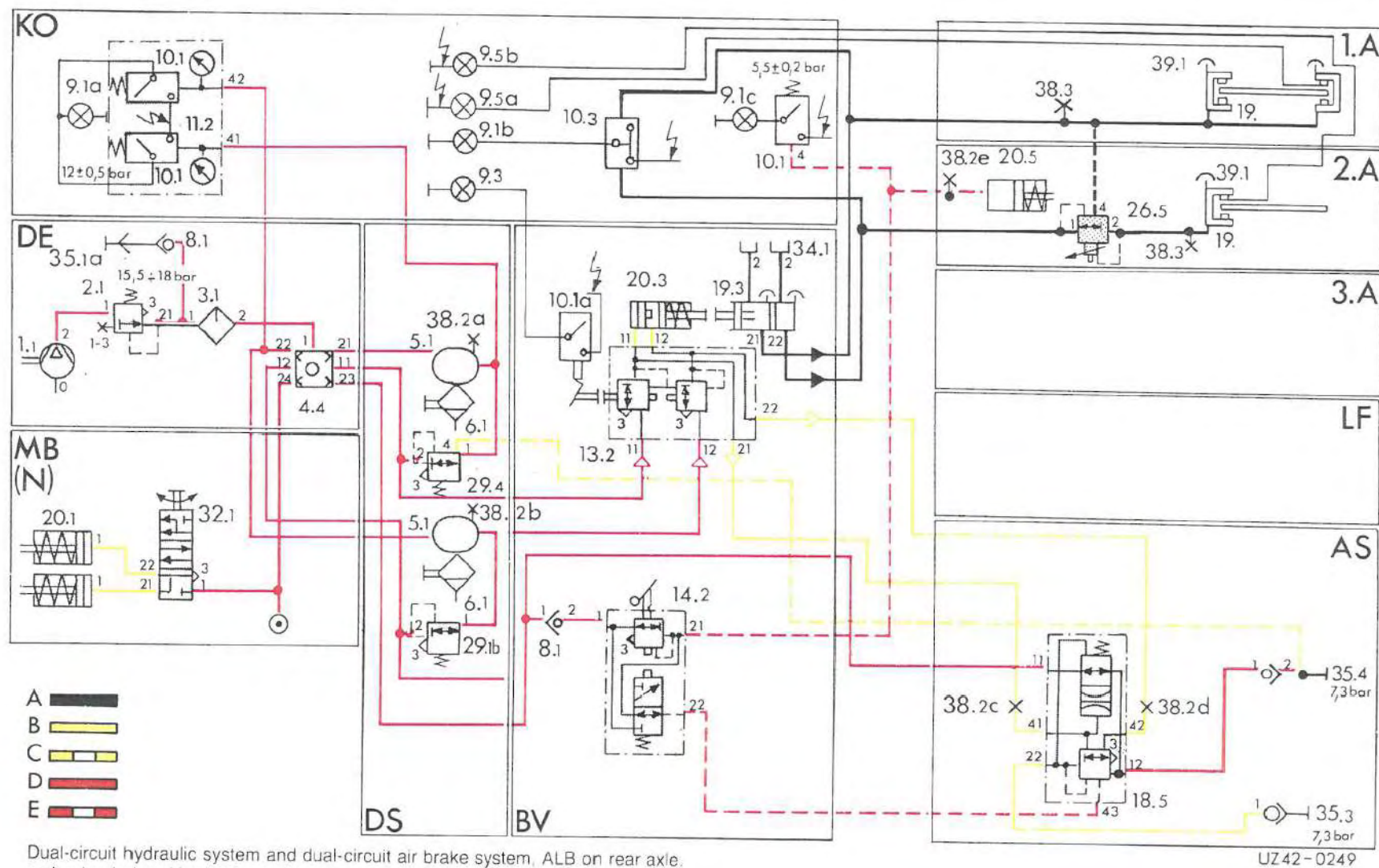
Dual-circuit hydraulic system and dual-circuit air brake system, ALB on rear axle, spring-loaded parking brake and two-line trailer brake system. From chassis end No. 084 389

Key to 1.7/3

- A Hydraulics
- B Brake pressure
- C Brake control pressure
- D Supply pressure
- E Supply control pressure

- 1.1 Air compressor
- 2.1 Pressure regulator with tyre inflating port
- 3.1 Antifreeze unit
- 4.4 Four-circuit protection valve
- 5.1 Air tank circuit 1
- 5.1a Air tank circuit 2
- 6.1 Drain valve
- 8.1 Non-return valve
- 9.1a Warning lamp supply pressure
- 9.1b Indicator lamp pressure difference
- 9.1c Indicator lamp parking brake
- 9.3 Stoplight
- 9.5a Indicator lamp wear indicator FA
- 9.5b Indicator lamp wear indicator RA
- 10.1 Electrical pressure switch supply pressure
- 10.1a Electrical pressure switch stoplight
- 10.3 Electrical pressure difference warning switch
- 11.2 Double pressure gauge
- 13.2 Service brake valve (dual-circuit)
- 14.2 Parking brake valve with brake test point
- 18.4 Trailer control valve (two-line) with breakaway valve
- 19. Fixed caliper
- 19.3 Master brake cylinder (dual-circuit)
- 20.1 Piston cylinder 4-wheel drive differential lock
- 20.3 Preload cylinder (dual-circuit)
- 20.5 Spring-loaded parking brake
- 26.5 ALB modulator
- 29.1 Pressure reducing valve
- 29.4 Pressure reducing valve with pneumatic control
- 32.1 Switchover valve 4-wheel differential lock
- 34.1 Equalizing tank
- 35.1 Coupling head brake
- 35.1a Coupling head front charging line
- 35.4 Coupling head supply
- 38.2a Test connection circuit 1
- 38.2b Test connection circuit 2
- 38.2c Test connection circuit 1
- 38.2d Test connection circuit 2
- 38.2e Test connection, spring-loaded cylinder
- 38.3 Test connection ALB modulator
- 39.1 Bleeder valve fixed caliper

Functional Diagram (Special Vehicle 435.115)



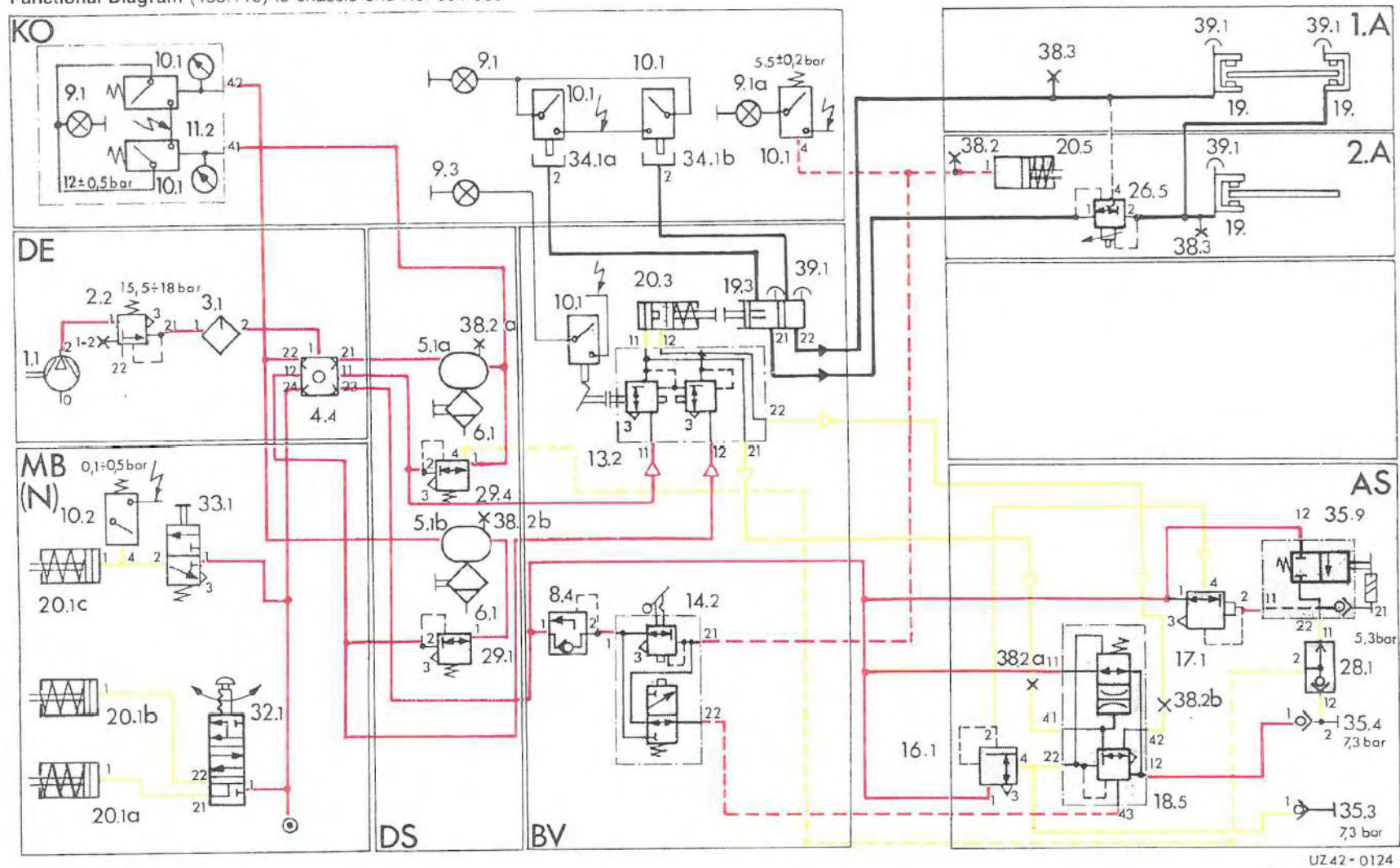
Dual-circuit hydraulic system and dual-circuit air brake system, ALB on rear axle, spring-loaded parking brake and two-line trailer brake system.
From chassis end No. 105 689

Key to 1.7/5

- A Hydraulics
- B Brake pressure
- C Brake control pressure
- D Supply pressure
- E Supply control pressure

- 1.1 Air compressor
- 2.1 Pressure regulator with tyre inflating port
- 3.1 Antifreeze unit
- 4.4 Four-circuit protection valve
- 5.1 Air tank circuit 1
- 5.1a Air tank circuit 2
- 6.1 Drain valve
- 8.1 Non-return valve
- 9.1a Warning lamp supply pressure
- 9.1b Indicator lamp pressure difference
- 9.1c Indicator lamp parking brake
- 9.3 Stoplight
- 9.5a Indicator lamp wear indicator FA
- 9.5b Indicator lamp wear indicator RA
- 10.1 Electrical pressure switch supply pressure
- 10.1a Electrical pressure switch stoplight
- 10.3 Electrical pressure difference warning switch
- 11.2 Double pressure gauge
- 13.2 Service brake valve (dual-circuit)
- 14.2 Parking brake valve with brake test point
- 18.4 Trailer control valve (2-line) with breakaway valve
- 19. Fixed caliper
- 19.3 Master brake cylinder (dual-circuit)
- 20.1 Piston cylinder, 4-wheel drive differential lock
- 20.3 Preload cylinder (dual-circuit)
- 20.5 Spring-loaded parking brake
- 26.5 ALB regulator
- 29.1 Pressure reducing valve
- 29.4 Pressure reducing valve with pneumatic control
- 32.1 Switchover valve 4-wheel differential lock
- 34.1 Equalizing tank
- 35.1 Coupling head brake
- 35.1a Front charging line
- 35.4 Coupling head supply
- 38.2a Test connection, circuit 1
- 38.2b Test connection, circuit 2
- 38.2c Test connection, circuit 1
- 38.2d Test connection, circuit 2
- 38.2e Test connection, spring loaded cylinder
- 38.3 Test connection ALB modulator
- 39.1 Bleeder valve, fixed caliper

Functional Diagram (435.115) to chassis end No. 084 388



A B C D E F

Dual-circuit hydraulic system and dual-circuit air brake system, ALB on rear axle and 1/2 front axle spring-loaded parking brake, single and dual-line trailer brake system.

Key to 1.7/7

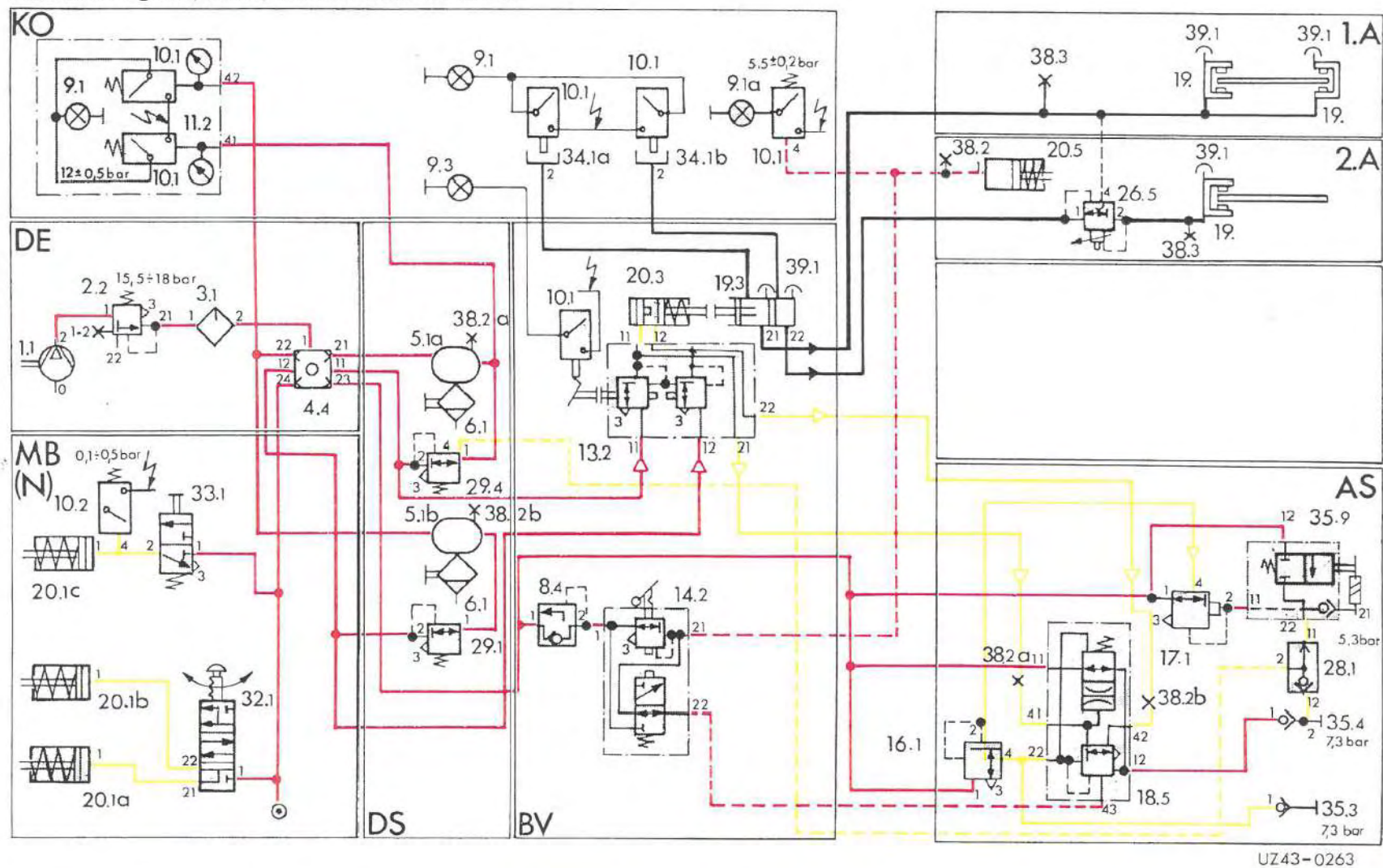
A Hydraulics
B Hydraulic control pressure
C Brake pressure
D Control pressure
E Supply
F Supply control pressure

1.A = 1st axle
2.A = 2nd axle
AS = Trailer control
BV = Actuating valve
DS = Accumulator
MB(N) = Engine brake (auxiliary load)
DE = Pressure generation
KO = Control switch

1.1 Air compressor
2.2 Pressure regulator
3.1 Antifreeze unit
4.4 Four-circuit protection valve
5.1a Air tank circuit 1
5.1b Air tank circuit 2
6.1 Water separator
8.4 Non-return valve with limited return flow
9.1 Warning lamp, red hydraulics
9.1a Warning lamp, red parking brake
9.3 Stoplight
10.1 Electrical pressure switch
10.2 Electrical pressure switch
11.2 Double pressure gauge
13.2 Service brake valve
14.2 Parking brake valve
16.1 Relay valve
17.1 Trailer control valve (1-line)
18.5 Trailer control valve (2-line)
19. Fixed caliper
19.3 Tandem master brake cylinder
20.1a Piston cylinder 4-wheel drive
20.1b Piston cylinder differential lock
20.1c Piston cylinder engine brake

20.3 Preload cylinder
20.5 Spring-loaded parking brake
26.5 ALB modulator
28.1 Two-way valve trailer operation
29.1 Pressure reducing valve
29.4 Pressure reducing valve, switchover
32.1 Switchover valve, 4-wheel drive and differential lock
33.1 Switchover valve engine brake
34.1a Brake fluid reservoir circuit 1
34.1b Brake fluid reservoir circuit 2
35.3 Coupling head brake (2-line)
35.4 Coupling head supply (2-line)
35.9 Coupling head (single line)
38.2 Test connection spring-loaded cylinder
38.2a Test connection circuit 1
38.2b Test connection circuit 2
38.3 Test connection ALB modulator
39.1 Vent

Functional Diagram (435.115) from chassis end No. 084 389

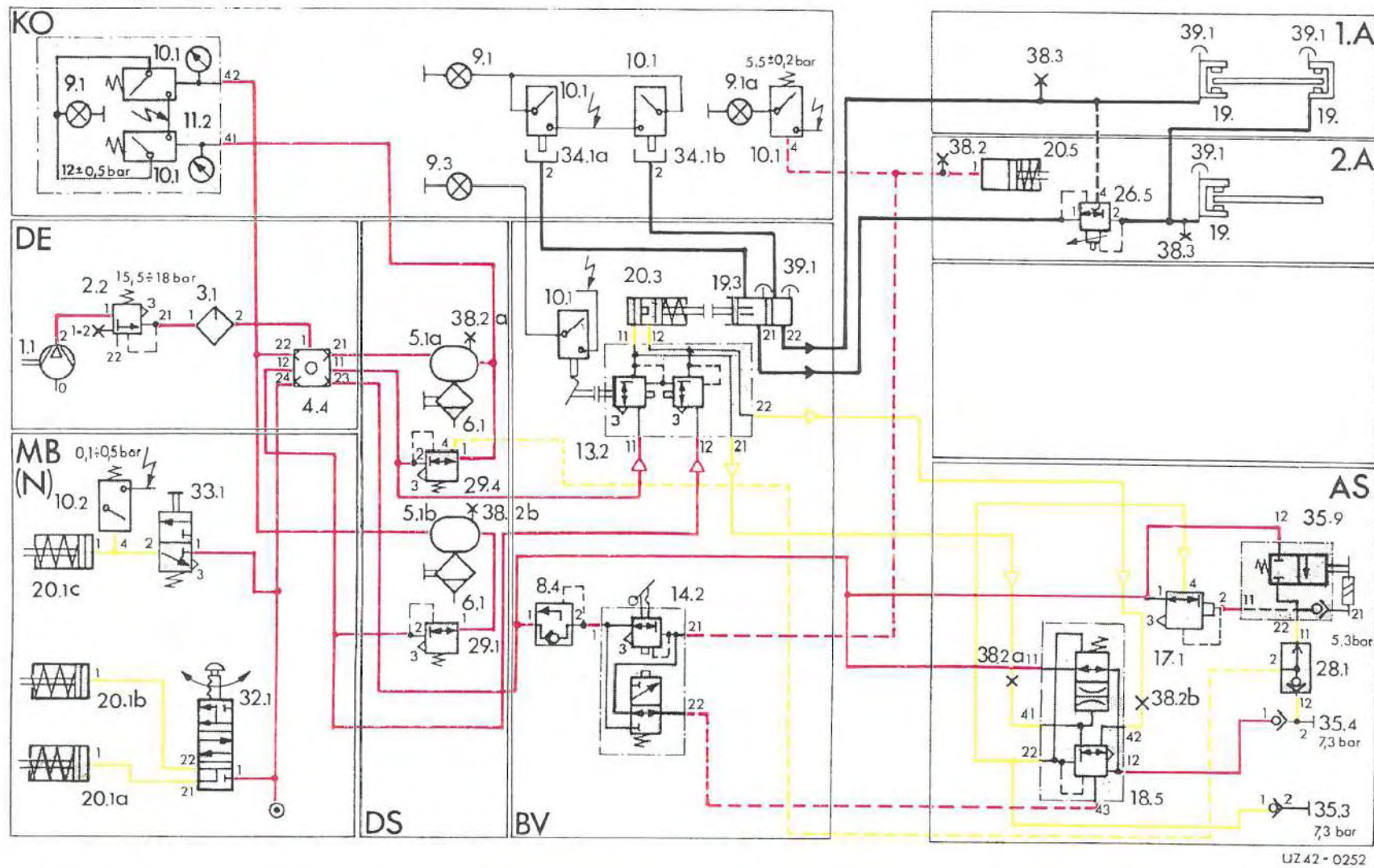


Dual-circuit hydraulic system and dual-circuit air brake system, ALB on rear axle, spring-loaded parking brake, single and dual-circuit trailer brake system.

Key to 1.7/9

A	Hydraulics	1.A	= 1st axle
B	Hydraulic control pressure	2.A	= 2nd axle
C	Brake pressure	AS	= Trailer control
D	Control pressure	BV	= Actuating valve
E	Supply	DS	= Accumulator
F	Supply control pressure	MB(N)	= Engine brake (auxiliary load)
		DE	= Pressure generation
		KO	= Control switch
1.1	Air compressor	20.3	Preload cylinder
2.2	Pressure regulator	20.5	Spring-loaded parking brake
3.1	Antifreeze unit	26.5	ALB modulator
4.4	Four-circuit protection valve	28.1	Two-way valve trailer operation
5.1a	Air tank circuit 1	29.1	Pressure reducing valve, switchover
5.1b	Air tank circuit 2	29.4	
6.1	Water separator	32.1	Switchover valve, 4-wheel drive and differential lock
8.4	Non-return valve with limited return flow	33.1	Switchover valve, engine brake
9.1	Warning lamp, red hydraulics	34.1 a	Brake fluid reservoir, circuit 1
9.1a	Warning lamp, red parking brake	34.1 b	Brake fluid reservoir, circuit 2
9.3	Stoplight	35.3	Coupling head brake, (2-line)
10.1	Electrical pressure switch	35.4	Coupling head supply (2-line)
10.2	Electrical pressure switch	35.9	Coupling head (single-line)
13.2	Service brake valve	38.2	Test connection spring-loaded cylinder
14.2	Parking brake valve	38.2a	Test connection circuit 1
16.1	Relay valve	38.2b	Test connection circuit 2
17.1	Trailer control valve (1-line)	38.3	Test connection ALB modulator
18.5	Trailer control valve (2-line)	39.1	Vent
19	Fixed caliper		
19.3	Tandem master brake cylinder		
20.1a	Piston cylinder 4-wheel drive		
20.1b	Piston cylinder differential lock		
20.1c	Piston cylinder engine brake		

Funktional Diagram (435.117) to chassis end No. 071 795



UJ 42 - 0252



Dual-circuit hydraulic system and dual-circuit air brake system, ALB on rear axle and $\frac{1}{2}$ front axle, spring-loaded parking brake, single and dual circuit trailer brake system

Key to 1.7/11

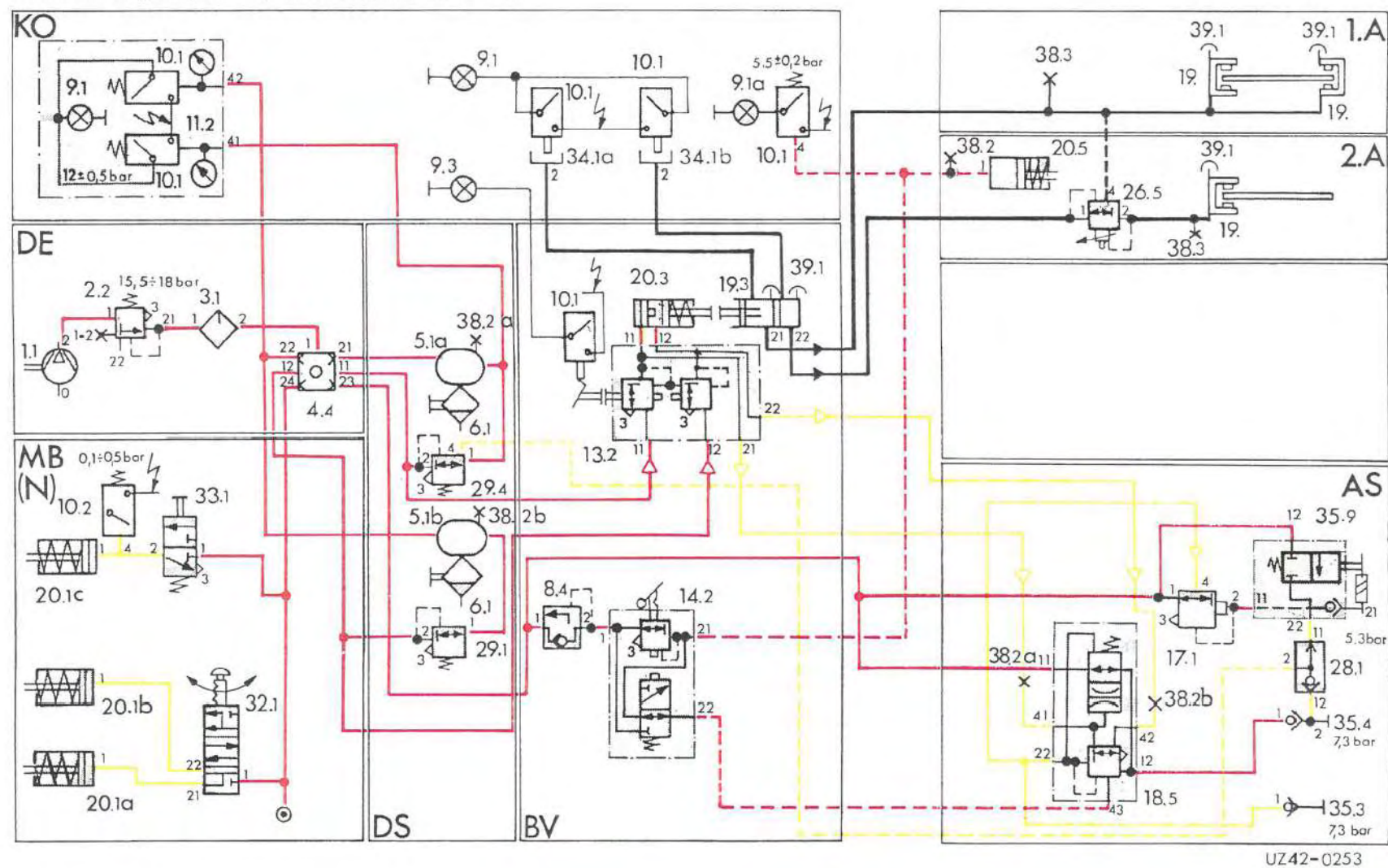
A Hydraulics
B Hydraulic control pressure
C Brake pressure
D Control pressure
E Supply
F Supply control pressure

1.A = 1st axle
2.A = 2nd axle
AS = Trailer control
BV = Actuating valve
DS = Accumulator
MB(N) = Engine brake (auxiliary load)
DE = Pressure generation
KO = Control switch

1.1 Air compressor
2.2 Pressure regulator
3.1 Antifreeze unit
4.4 Four-circuit protection valve
5.1a Air tank circuit 1
5.1b Air tank circuit 2
6.1 Water separator
8.4 Non-return valve with limited return flow
9.1 Warning lamp, red hydraulics
9.1a Warning lamp, red parking brake
9.3 Stoplight
10.1 Electrical pressure switch
10.2 Electrical pressure switch
11.2 Double pressure gauge
13.2 Service brake valve
14.2 Parking brake valve
17.1 Trailer control valve (1-line)
18.5 Trailer control valve (2-line)
19 Fixed caliper
19.3 Tandem master brake cylinder
20.1a Piston cylinder 4-wheel drive
20.1b Piston cylinder differential look
20.1c Piston cylinder engine brake

20.3 Preload cylinder
20.5 Spring-loaded parking brake
26.5 ALB modulator
28.1 Two-way valve trailer operation
29.1 Pressure reducing valve
29.4 Pressure reducing valve, switchover
32.1 Switchover valve, 4-wheel drive and differential look
33.1 Switchover valve engine brake
34.1a Brake fluid reservoir circuit 1
34.1b Brake fluid reservoir circuit 2
35.3 Coupling head brake (2-line)
35.4 Coupling head supply (2-line)
35.9 Coupling head (single-line)
38.2 Test connection spring-loaded cylinder
38.2a Test connection circuit 1
38.2b Test connection circuit 2
38.3 Test connection ALB modulator
39.1 Vent

Functional Diagram (435.117) from chassis end No. 071 796



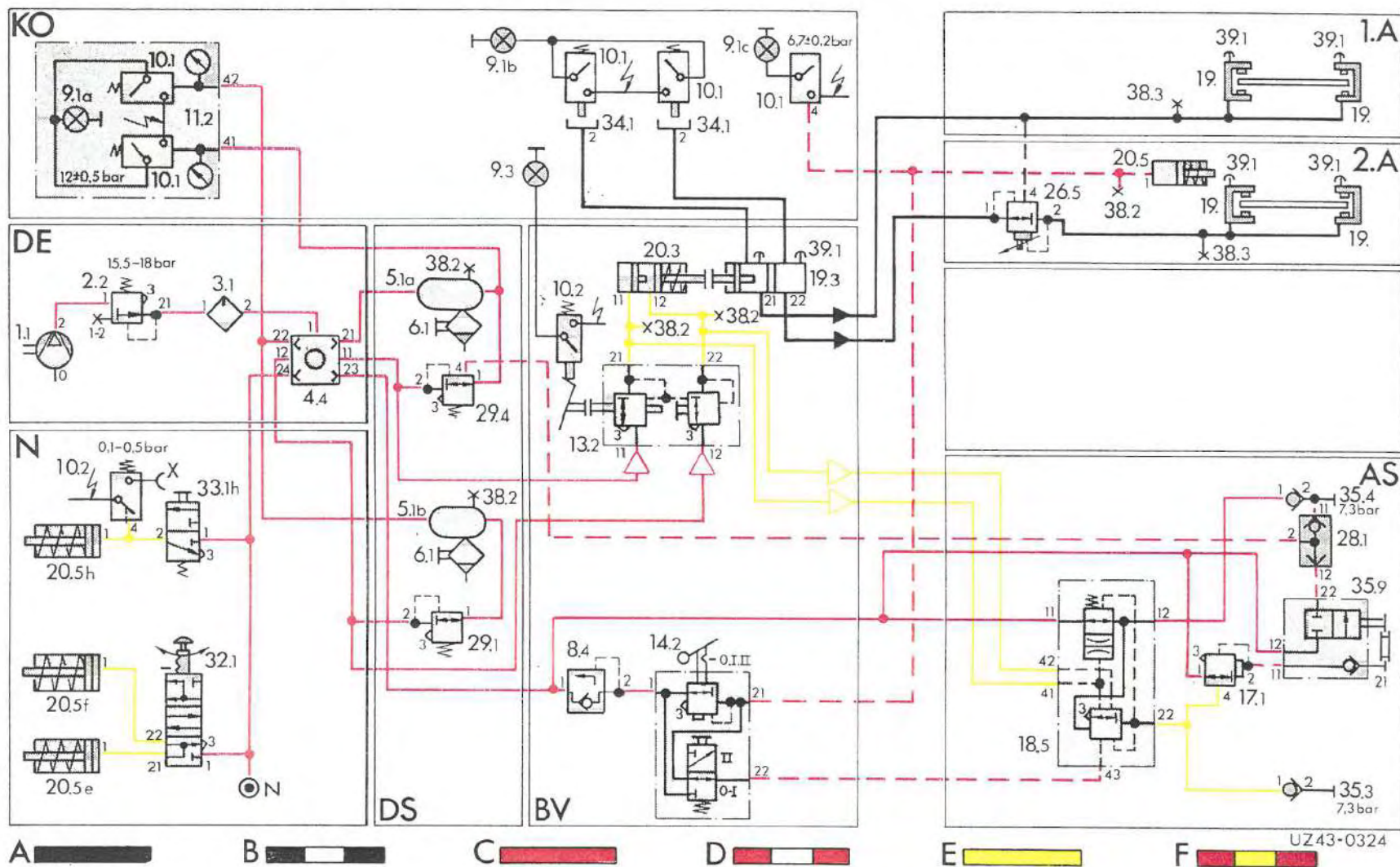
A B C D E F

Dual-circuit hydraulic system and dual circuit air-brake system, ALB on rear axle, spring-loaded parking brake, single and dual-circuit trailer brake system

Key to 1.7/13

A	Hydraulics	1.A	= 1st axle
B	Hydraulic control pressure	2.A	= 2nd axle
C	Brake pressure	AS	= Trailer control
D	Control pressure	BV	= Actuating valve
E	Supply	DS	= Accumulator
F	Supply control pressure	MB(N)	= Engine brake (auxiliary load)
		DE	= Pressure generation
		KO	= Control switch
1.1	Air compressor	20.3	Preload cylinder
2.2	Pressure regulator	20.5	Spring-loaded parking brake
3.1	Antifreeze unit	26.5	ALB modulator
4.4	Four-circuit protection valve	28.1	Two-way valve trailer operation
5.1a	Air tank circuit 1	29.1	Pressure reducing valve
5.1b	Air tank circuit 2	29.4	Pressure reducing valve, switchover
6.1	Water separator	32.1	Switchover valve, 4-wheel drive and differential lock
8.4	Non-return valve with limited return flow	33.1	Switchover valve engine brake
9.1	Warning lamp, red hydraulics	34.1a	Brake fluid reservoir circuit 1
9.1a	Warning lamp, red parking brake	34.1b	Brake fluid reservoir circuit 2
9.3	Stoplight	35.3	Coupling head brake (2-line)
10.1	Electrical pressure switch	35.4	Coupling head supply (2-line)
10.2	Electrical pressure switch	35.9	Coupling head (single-line)
11.2	Double pressure gauge	38.2	Test connection spring-loaded cylinder
13.2	Service brake valve	38.2a	Test connection circuit 1
14.2	Parking brake valve	38.2b	Test connection circuit 2
17.1	Trailer control valve (1-line)	38.3	Test connection ALB modulator
18.5	Trailer control valve (2-line)	39.1	Vent
19	Fixed caliper		
19.3	Tandem master brake cylinder		
20.1a	Piston cylinder 4-wheel drive		
20.1b	Piston cylinder differential lock		
20.1c	Piston cylinder engine brake		

Functional Diagram (435.117 - SA 35 988)



Dual-circuit hydraulic system and dual-circuit air brake system, disk brake with 2nd pair of fixed calipers at rear axle, ALB at rear axle, combined single-line and dual-line trailer brake system

Key to 1.7/15

A	Hydraulics
B	Hydraulic control line
C	Supply
D	Supply control pressure
E	Working/pilot control pressure
F	Supply/brake pressure, single-line

KO	Control switch
DE	Pressure generator
MB (N)	Exhaust brake/auxiliary load
DS	Accumulator
BV	Actuating valves
1. A	1st axle
2. A	2nd axle
AS	Trailer control

1.1	Compressor
2.2	Pressure regulator
3.1	Antifreeze device
4.4	Four-circuit protection valve
5.1 a	Compressed air reservoir
5.1 b	Compressed air reservoir
6.1	Water separator
8.4	Non-return valve
9.1 a	Indicator lamp
9.1 b	Indicator lamp
9.1 c	Indicator lamp
9.3	Stop lamp
10.1	Off switch
11.2	Dual pressure gauge
13.2	Service brake valve
14.2	Parking brake valve
17.1	Trailer control valve
18.5	Trailer control valve
19.	Fixed caliper
19.3	Hydraulic master cylinder
20.3	Pneumatic piston cylinder
20.5	Spring brake cylinder
20.5 e	Spring brake cylinder
20.5 F	Spring brake cylinder
20.5 h	Spring brake cylinder
26.5	Automatic brake force regulator
28.1	Check valve
29.1	Pressure relief valve
29.4	Pressure relief valve
32.1	1/3-way valve
33.1 h	1/3-way valve
34.1	Tank
35.3	Coupling head
35.4	Coupling head
35.9	Coupling head
38.2	Test connection
38.3	Test connection
39.1	Bleeder point
X	Connection socket trailer (54 g)

Single-cylinder (air compressor)

Circuit 1

Circuit 2

with limited return flow

Supply pressure

Brake fluid level

Spring brake

dual-circuit

1-line

2-line, with integrated breakaway valve

dual-circuit

(Actuating cylinder - dual - circuit)

pulling

Four-wheel drive

Differential lock

Exhaust brake

Shuttle valve

with pressure safeguard

reversible with pressure safeguard

Four-wheel drive/differential lock

Exhaust brake

Hydraulic system

2-line, brake

2-line, supply

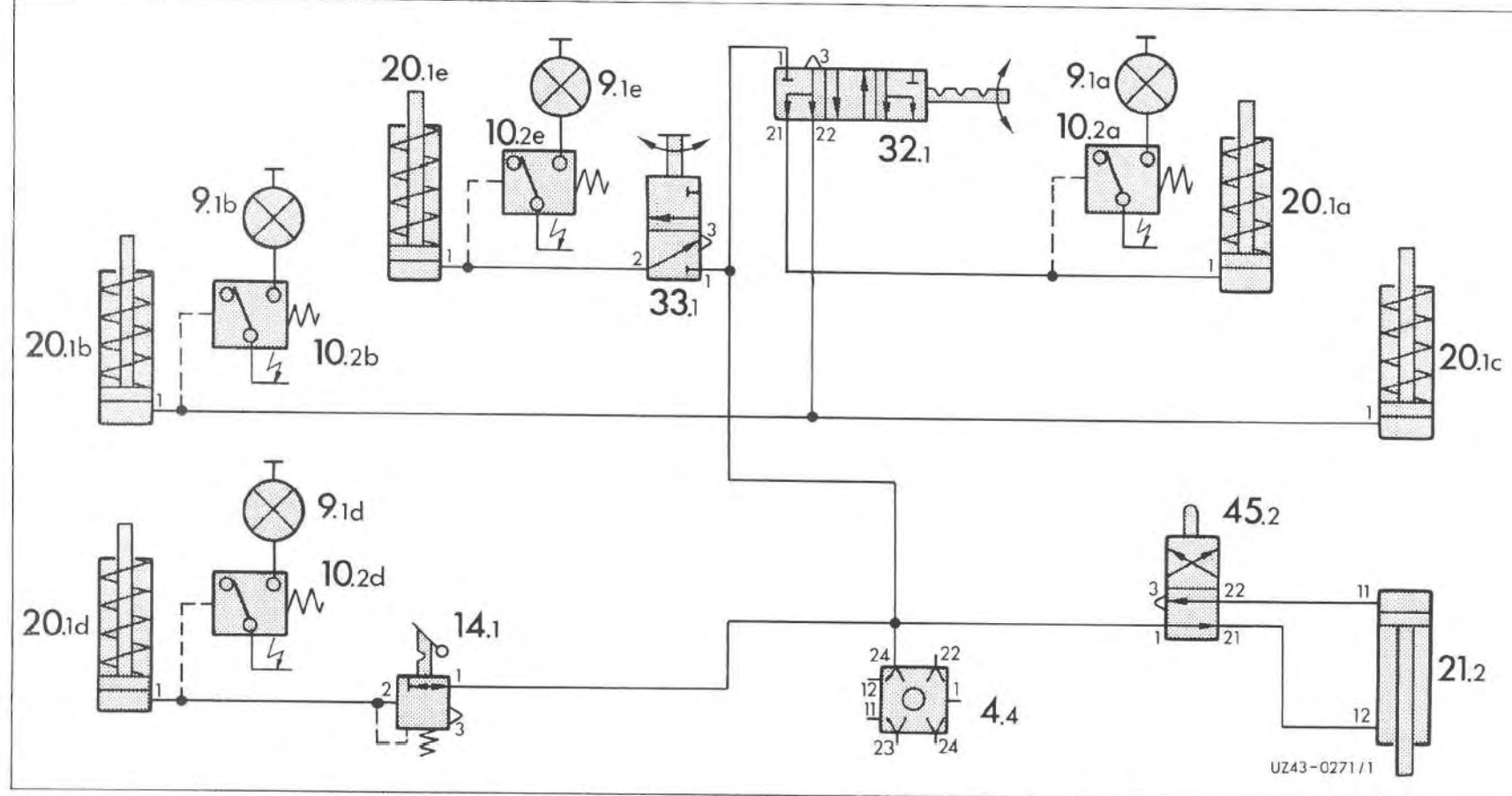
1-line, brake and 1/2-way valve

Pneumatic

Hydraulic

Hydraulic

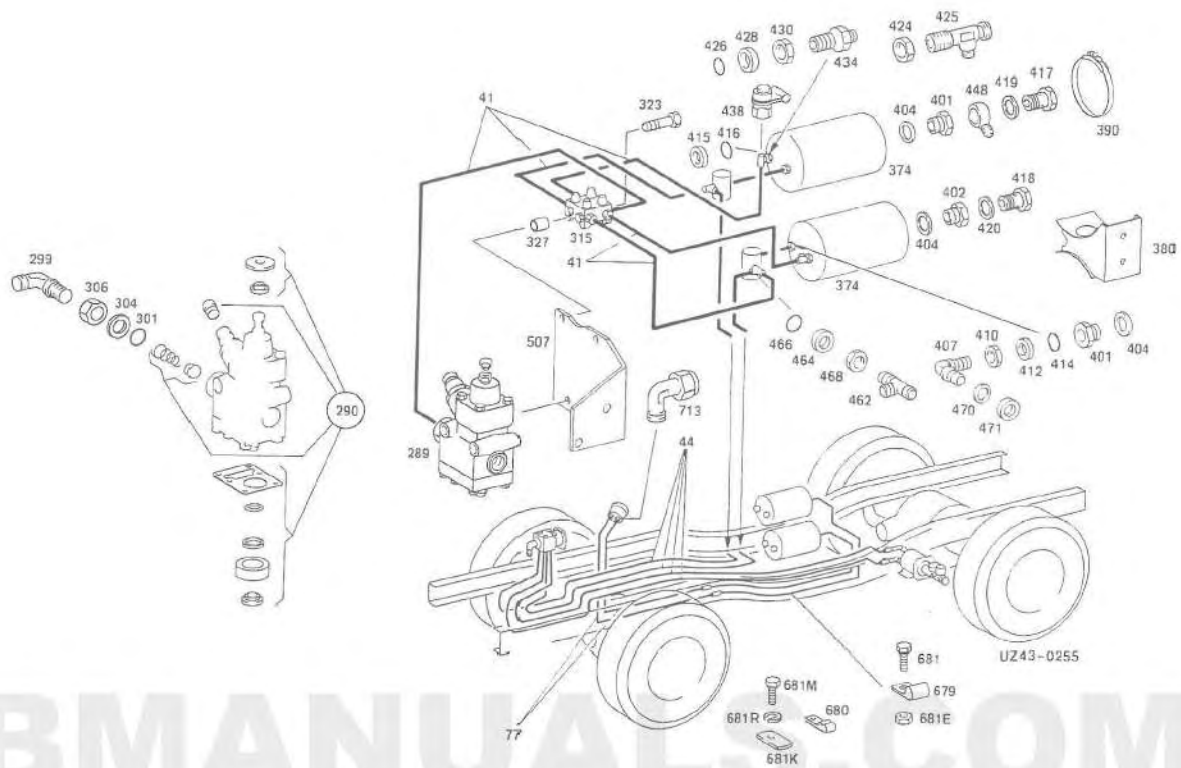
Functional Diagram Auxiliary Load (N) (435.115.117)



- | | | | | | |
|-------|--|-------|---|-------|---|
| 4.4 | Four-circuit protection valve | 10.2d | Electrical pressure switch PTO shaft | 20.1e | Piston cylinder power take-off |
| 9.1a | Indicator lamp 4-wheel drive | 10.2e | Electrical pressure switch power take-off | 21.2 | Piston cylinder double-acting planetary gearbox train |
| 9.1b | Indicator lamp differential lock | 14.1 | Switchover valve PTO shaft | 32.1 | Switchover valve 4-wheel drive differential lock |
| 9.1d | Indicator lamp PTO shaft | 20.1a | Piston cylinder 4-wheel drive | 33.1 | Switchover valve power take-off |
| 9.1e | Indicator lamp power take-off | 20.1b | Piston cylinder differential lock FA | 45.2 | Switchover valve planetary gear |
| 10.2a | Electrical pressure switch 4-wheel drive | 20.1c | Piston cylinder differential lock RA | | |
| 10.2b | Electrical pressure switch differential lock | 20.1d | Piston cylinder PTO shaft | | |

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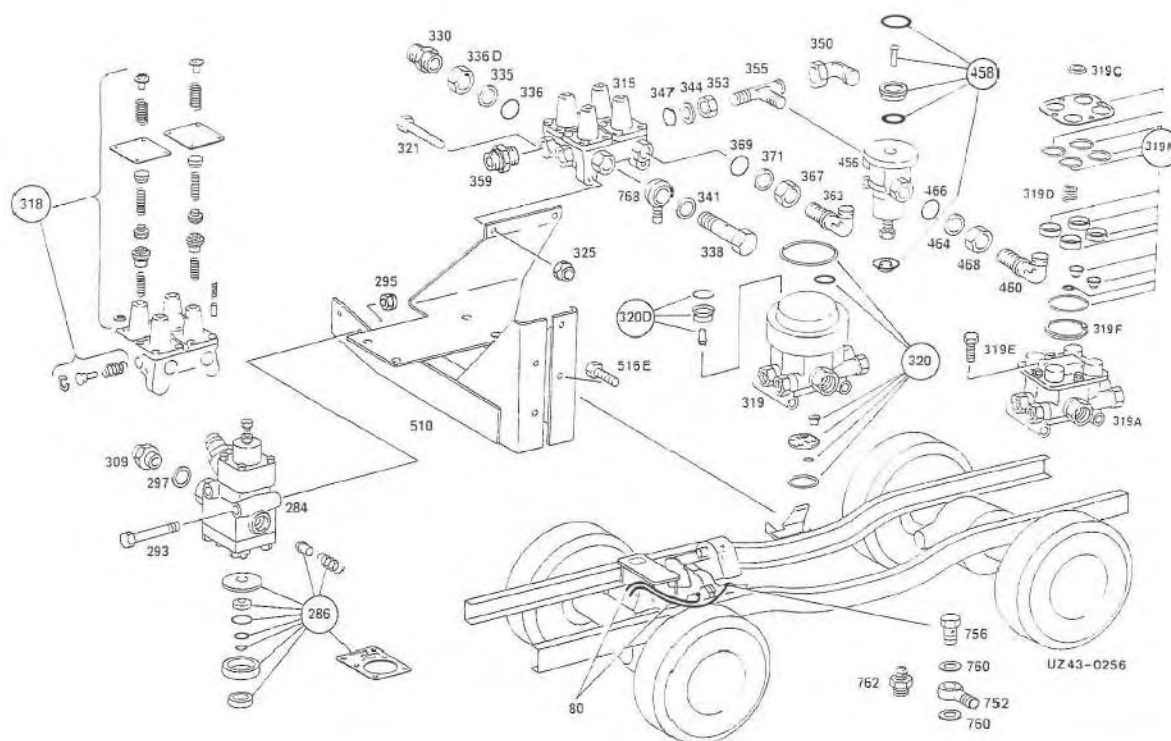
Exploded View



Brake System

41	Line	419	Sealing ring
44	Hose	420	Sealing ring
77	Hose	424	Nut
289	Pressure regulator	425	Pipe socket
290	Repair kit	426	Sealing ring
299	Pipe elbow	428	Thrust ring
301	Sealing ring	430	Nut
304	Thrust ring	434	Pipe socket
306	Nut	438	Test connection
315	Four-circuit protection valve	448	Inlet union
323	Screw	462	T-piece
327	Air tank	464	Thrust ring
380	Bracket	466	Sealing ring
390	Clip	468	Nut
401	Screw fitting	470	Thrust ring
402	Screw fitting	471	Sealing ring
404	Sealing ring	507	Bracket
407	Pipe elbow	679	Clip
410	Nut	680	Clip
412	Thrust ring	681	Screw
414	Sealing ring	681E	Nut
415	Thrust ring	681K	Gasket
416	Sealing ring	681M	Bolt
417	Banjo bolt	681R	Lock washer
418	Banjo bolt	713	Pipe socket

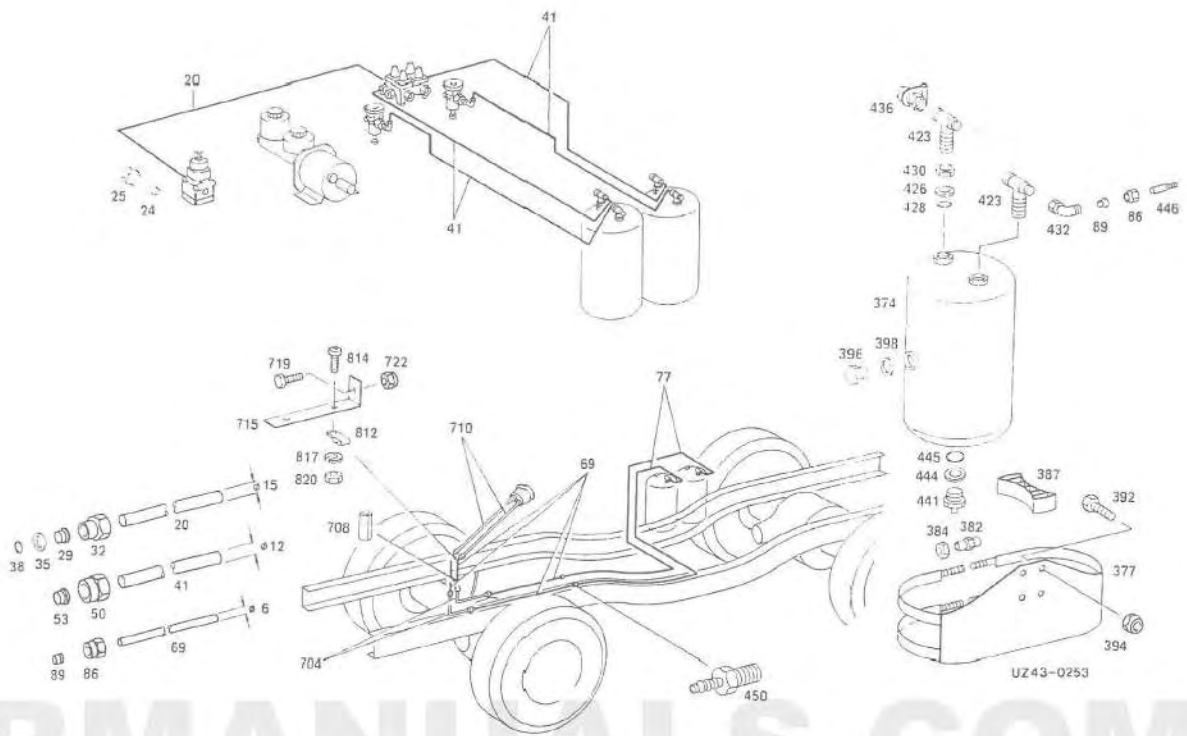
Exploded View



Brake System

80	Hose	341	Sealing ring
284	Pressure regulator	344	Thrust ring
286	Repair kit	347	Sealing ring
293	Bolt	350	Pipe socket
295	Nut	353	Nut
297	Sealing ring	355	Pipe socket
309	Double adapter	359	Threaded union
315	Four-circuit protection valve	363	Pipe socket
318	Repair kit	367	Nut
319	Four-circuit protection valve	369	Sealing ring
319A	Casing	371	Thrust ring
319C	Washer	456	Pressure limiting valve
319D	Spring	458	Repair kit
319E	Bolt	460	Pipe socket
319F	Retaining ring	464	Thrust ring
319M	Repair kit	466	Sealing ring
320	Repair kit, seals	468	Nut
320D	Repair kit, sleeves	510	Bracket
321	Bolt	516E	Bolt
325	Nut	752	Inlet union
330	Double adapter	756	Banjo bolt
335	Thrust ring	760	Sealing ring
336	Sealing ring	762	Pipe socket
336D	Nut	768	Inlet union
338	Banjo bolt		

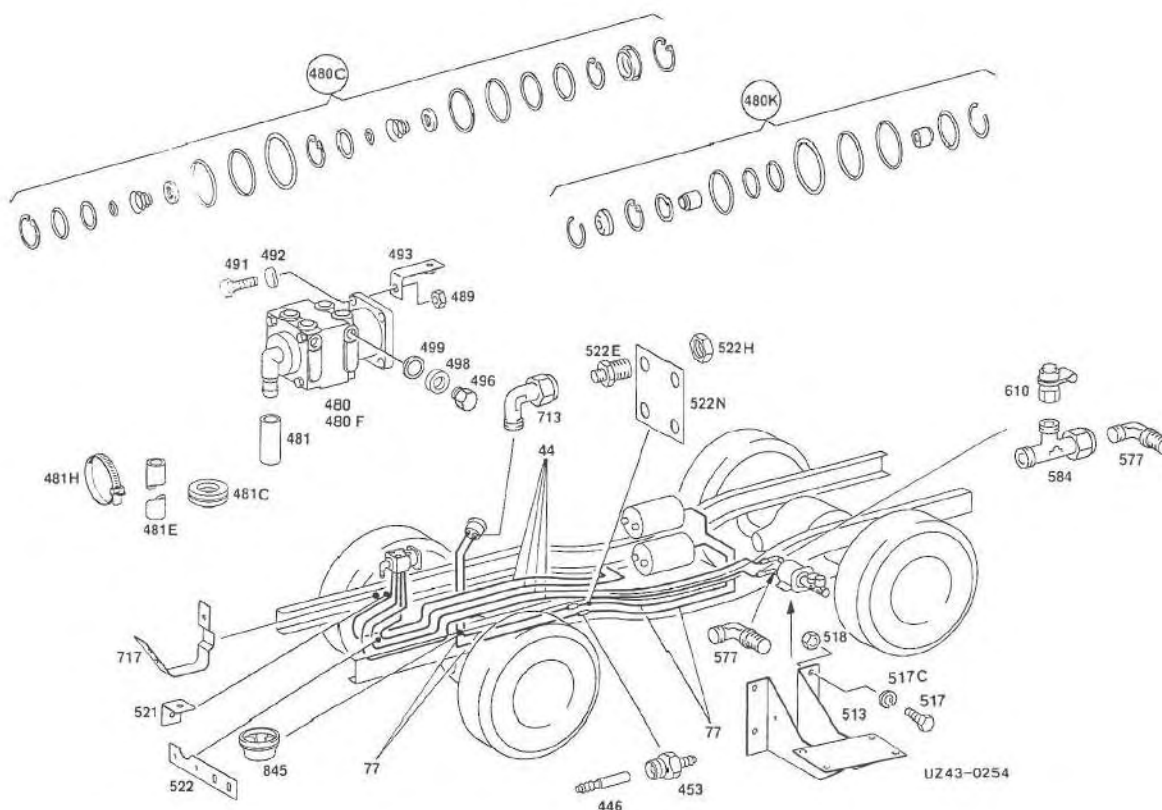
Exploded View



Air tank

20	Pipe	398	Sealing ring
24	Bite ring	423	T-piece
25	Union nut	426	Sealing ring
29	Bite ring	428	Thrust ring
32	Union nut	430	Nut
35	Thrust ring	432	Pipe socket
38	Sealing ring	436	Test connection
41	Line	441	Drain valve
50	Union nut	444	Thrust ring
53	Bite ring	445	Sealing ring
69	Line	446	Pipe socket
77	Hose	450	Pipe socket
86	Union nut	704	Pipe
89	Bite ring	708	Hose
374	Air tank	710	Brake hose
377	Bracket	715	Bracket
382	Turnbuckle nut	719	Bolt
384	Nut	722	Nut
387	Shim	812	Clip
392	Bolt	814	Bolt
394	Nut	817	Lock washer
396	Screw plug	820	Nut

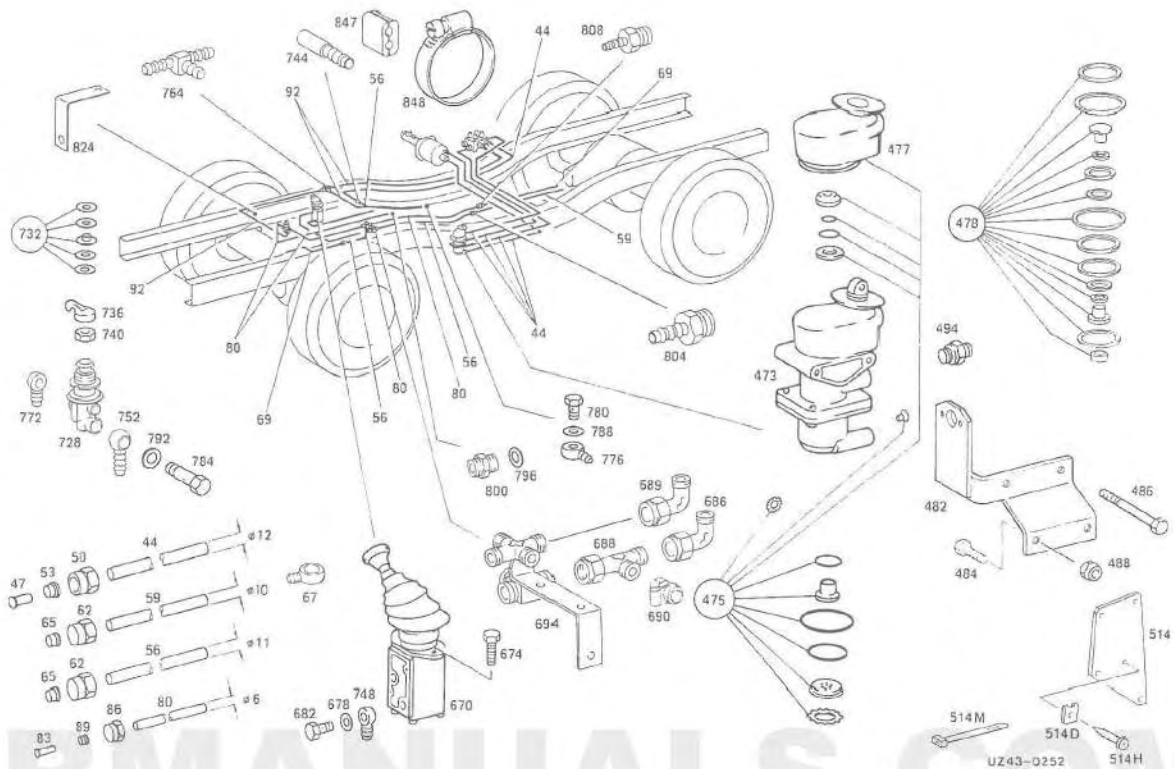
Exploded view



Brake System

44	Hose	498	Thrust ring
77	Hose	499	Sealing ring
446	Pipe socket	513	Bracket
453	Pipe socket	517	Bolt
480	Brake valve, tractor, optional	517C	Lock washer
480C	Repair kit	518	Nut
480F	Brake valve, tractor, optional	521	Bracket
480K	Repair kit	522	Bracket
481	Hose	522E	Pipe socket
481C	Sleeve	522H	Nut
481E	Rubber sleeve	522N	Shim
481H	Clip	577	Pipe elbow
489	Nut	584	Pipe socket
491	Bolt	610	Test connection
492	Spring washer	713	Pipe socket
493	Bracket	717	Bracket
496	Screw plug	845	Sleeve

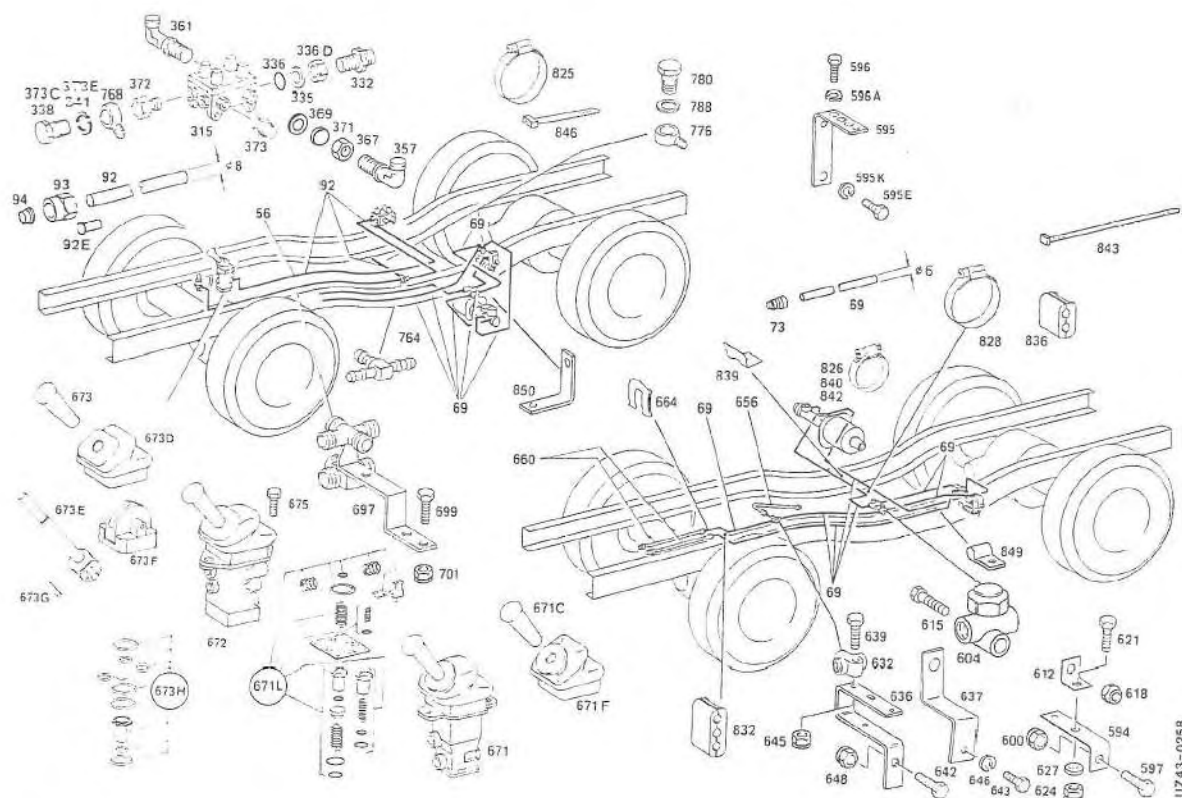
Exploded View



Break System

44	Hose	514H	Screw	796	Sealing ring
47	Sleeve	514M	Clip	800	Pipe socket
50	Union nut	670	Parking brake valve	804	Pipe socket
53	Bite ring	674	Bolt	808	Pipe socket
56	Hose	678	Sealing ring	824	Bracket
59	Line	682	Banjo bolt	847	Gasket
62	Union nut	686	Pipe socket	848	Clip
65	Bite ring	688	Pipe socket		
67	Line	689	Pipe socket		
69	Line	690	Test connection		
80	Hose	694	Bracket		
83	Sleeve	728	Compressed air valve- differential lock		
86	Union nut	732	Repair kit		
89	Bite ring	736	Knob		
92	Hose	740	Nut		
473	Brake valve	744	Pipe socket		
475	Repair kit	748	Inlet union		
477	Dust cap	752	Inlet union		
478	Repair kit	764	Connection piece		
482	Brackets	772	Inlet union		
484	Bolt	776	Inlet union		
486	Bolt	780	Banjo bolt		
488	Nut	784	Banjo bolt		
494	Threaded union	788	Sealing ring		
514	Bracket	792	Sealing ring		
514D	Retainer				

Exploded View



Brake System

56	Hose	597	Bolt	673	Grip
69	Line	600	Nut	673D	Sleeve
73	Union screw	604	Test connection	673E	Actuating unit
92	Hose	612	Bracket	673F	Cover
92E	Sleeve	615	Bolt	673G	Pin
93	Union nut	618	Nut	673H	Repair kit
94	Bite ring	621	Bolt	675	Bolt
315	Four-circuit protection valve	624	Nut	697	Bracket
332	Pipe socket	627	Washer	699	Bolt
335	Thrust ring	632	Distributor	701	Nut
336	Sealing ring	636	Bracket	764	Connection piece
336D	Nut	637	Bracket	768	Inlet union
338	Banjo bolt	639	Bolt	776	Inlet union
341	Sealing ring	642	Bolt	780	Banjo bolt
357	Pipe socket	643	Bolt	788	Sealing ring
351	Pipe socket	645	Nut	825	Clip
367	Nut	646	Lock washer	826	Clip
369	Sealing ring	648	Nut	828	Clip
371	Thrust ring	656	Brake hose	832	Mount
372	Threaded piece	660	Brake hose	836	Mount
373	Threaded union	664	Retainer	839	Bracket
373C	Banjo bolt	671	Parking brake valve, optional	840	Clip
373E	Sealing ring	671C	Grip	842	Clip
594	Bracket	671F	Sleeve	843	Clip
595	Bracket	671L	Repair kit	846	Clip
595E	Bolt	672	Parking brake valve, optional	849	Clip
595K	Lock washer			850	Bracket
596	Bolt				
596A	Lock washer				